

MANN 192311090

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SIMATS Online Programming Lab
JAVA PROGRAMMING

MANN 192311090RB

QUESTIONS

Q1 Student Registration System 10 marks ✓

Q2 E-Commerce Cart System 10 marks ✓

Q3 Library Management System 10 marks ✓

Q4 Employee Data Processing 10 marks ✓

Q5 Online Voting System 10 marks ✓

5/5 answered

Q2 E-Commerce Cart System 10 marks

Suppose that an online shopping system maintains a cart where products are added, removed, and displayed in order.
(a) List three Collection classes applicable for cart management.
(b) Represent how cart items are stored using a List-based structure.
(c) Write a Java program using ArrayList and Iterator to:

- Add items
- Remove an item
- Display all items

Your Code

```
1 import java.util.ArrayList;
2 import java.util.Iterator;
3
4 public class Main {
5     public static void main(String[] args) {
6         ArrayList<String> cart = new ArrayList<>();
7         cart.add("Milk");
8         cart.add("Bread");
9         cart.add("Butter");
10
11         Iterator<String> it = cart.iterator();
12         while (it.hasNext()) {
13             String item = it.next();
```

Input
stdin input

Output



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Q3 Library Management System 10 marks

Suppose that a library stores books with unique ISBN numbers and needs fast retrieval.
(a) Name three Map implementations in Java.
(b) Design a schema-like representation using Map for storing ISBN and book details.
(c) Write a Java program using HashMap to:

- Insert book details
- Retrieve a book
- Display all books

Your Code

```
1 import java.util.HashMap;
2 public class Main {
3     public static void main(String[] args) {
4         HashMap<String, String> books = new HashMap<>();
5         books.put("978-1-234-56789-7", "Java Basics");
6         books.put("978-1-4028-9462-6", "Data Structures");
7         books.put("978-0-306-40615-7", "Algorithms");
8         String isbn = "978-1-234-56789-7";
9         String book = books.get(isbn);
10        System.out.println("Retrieved book for ISBN " + isbn + ": " + book);
11        System.out.println("All books:");
12        for (String key : books.keySet()) {
13            System.out.println("ISBN: " + key + " Book: " + books.get(key));
```

Input
stdin input

Output



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5/5 answered

Q4 Employee Data Processing10 marks

Suppose that a company processes employee records and filters employees based on salary.

(a) List three advantages of Generics in Collections.

(b) Represent how employee data is stored using List with Generics.

(c) Write a Java program using Iterator to:

- Store employee objects
- Display employees with salary > 50,000

Your Code

```
1 import java.util.*;
2
3 public class Main {
4     public static void main(String[] args) {
5         List<Employee> emplist = new ArrayList<Employee>();
6         emplist.add(new Employee(1, "Asha", 60000));
7         emplist.add(new Employee(2, "Ravi", 45000));
8         emplist.add(new Employee(3, "Meena", 75000));
9         emplist.add(new Employee(4, "Karthik", 50000));
10        Iterator<Employee> it = emplist.iterator();
11        while (it.hasNext()) {
12            Employee e = it.next();
13            if (e.getSalary() > 50000) {
14                System.out.println(e.getId() + " " + e.getName() + " ");
15            }
16        }
17    }
18 }
```

Input

stdin input

Output



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10 marks

5/5 answered

Q1 Student Registration System10 marks

Suppose that a university system stores student data, where each student must be unique and can enroll in multiple courses.

(a) Enumerate three Collection interfaces suitable for this system.

(b) Design a Collection structure (diagram/representation) to store students and their enrolled courses.

(c) Write a Java program using HashSet and HashMap with Generics to:

- Add students
- Map students to courses
- Display data using Iterator

Your Code

```
1 import java.util.*;
2
3 public class Main {
4     public static void main(String[] args) {
5         HashSet<Student> set = new HashSet<>();
6         HashMap<Student, HashSet<String>> map = new HashMap<>();
7         Student s1 = new Student("S1", "MANN");
8         Student s2 = new Student("S2", "RAVI");
9         set.add(s1);
10        set.add(s2);
11        HashSet<String> c1 = new HashSet<>();
12        c1.add("Java");
13        c1.add("DBMS");
14        HashSet<String> c2 = new HashSet<>();
15        c2.add("Python");
16        map.put(s1, c1);
17        map.put(s2, c2);
18    }
19 }
```

Input

stdin input

Output